

Rego 600 / 63x Hardware Reference Manual

A comprehensive synthesis of the reverse-engineered 80C51 platform architectures, timings, and I/O structures.

1. Core CPU & Temporal Framework

Clock Frequency: 11.0592 MHz external crystal oscillator. In the standard 80C51 architecture, this frequency is divided by 12, yielding an internal machine cycle clock speed of **921,600 Hz** (921.6 machine cycles per millisecond).

Main Loop Frequency: The main system control loop executes at exactly **5 Hz** (200 ms per complete cycle). This execution timing is strictly managed via internal 80C51 timers to ensure predictable processing of thermal metrics.

2. Software-Based Phase Existence & Sequence Detection

Existence Verification: The firmware implements a 150-cycle software polling loop utilizing a logical OR accumulative strategy. If a phase fails to pulse high even once during this 200 ms window, a fault is registered.

Phase Rotation Sequence (L1 → L2 → L3): The true rotational sequence is verified inside a highly critical polling routine (FUN_CODE_87be) that temporarily disables global interrupts (CLR EA). To prevent continuous browser or simulator freezing while maintaining perfect 50 Hz real-world realism, the phase states must shift every **6,144 machine cycles** (calculated as 18,432 cycles per 20 ms period divided by 3 distinct operational phases).

Step	Active Phase	Hex Mask	Binary State (Bit 6, 5, 4)
0	L1 Only	0x10	0b00010000
1	L2 Only	0x20	0b00100000
2	L3 Only	0x40	0b01000000

3. I/O Expansion, Analog Multiplexing & Display Subsystem

Digital Input Capture: Handled sequentially via a chain of CD4021 parallel-to-serial shift registers mapped to CPU pin P1.5_4021_DAT. Software bit-banging uses P4.5 (LATCH) and P4.7 (CLK).

Analog Sensor Sampling: Relies on a combination of CD4094 shift registers driving two CD4051 8-channel analog multiplexers. To ensure maximum noise immunity and prevent signal cross-contamination during highly sensitive ADC conversions, the CPU's PWM output is driven to its maximum capacity, locking the address lines in a stable state.

Display Interface: Employs an isolated I2C master configuration connecting the 80C51 to a localized PIC16C63 slave controller. The primary microcontroller never renders individual pixels or handles complex timing loops; it outputs raw character arrays and custom segment indices directly to the PIC

processor, which hosts the physical hardware font maps and special icons (compressor, auxiliary heating elements).